Decision-Focused Learning for Kidney Exchange Programs

Why Does It Help, and Why Is the Gain Limited?

Weikang Fu

MILA
Université de Montréal

Final Internship Presentation

From the Previous Talk

First presentation

Question Can decision-focused learning improve controlled KEP decisions?

Finding Promising signal, but robustness remained unclear.

Final presentation

Question When does DFL help, fail, or become unnecessary?

Answer It is a candidate-reranking effect inside the feasible exchange landscape.

Main shift

From **method comparison** to **mechanism understanding**.

Recap: Kidney Exchange as a Weighted Decision Problem

Input

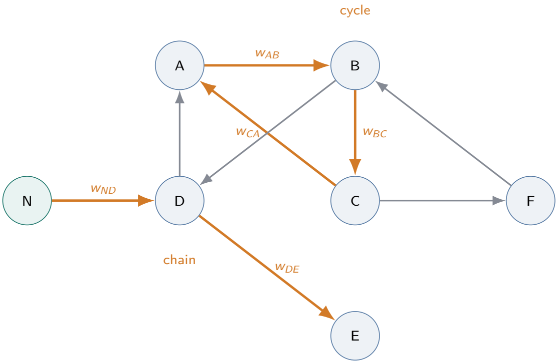
A directed compatibility graph with recipient–donor pairs and non-directed donors.

Decision

Select vertex-disjoint cycles and chains under length limits.

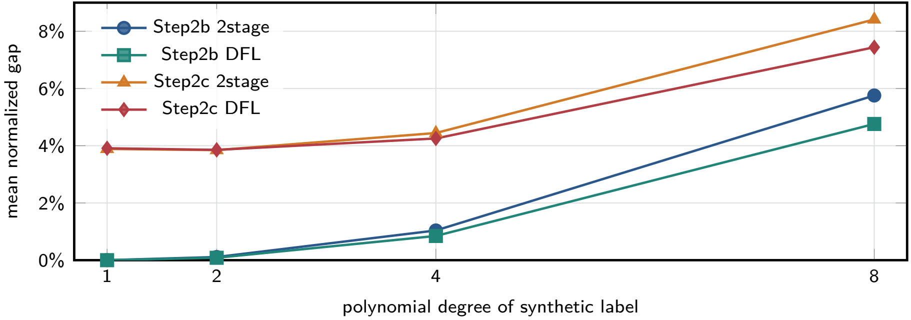
Objective

Maximize total predicted transplant benefit.



The learning model predicts edge weights; the optimizer chooses the exchange plan.

Main Experimental Backbone



Step2b

Noiseless polynomial labels: gap rises sharply as label degree increases.

Step2c

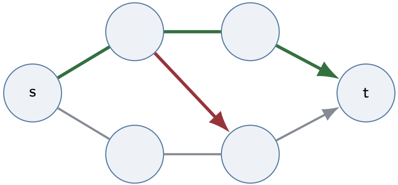
Multiplicative noise shifts the whole curve upward; DFL helps most at degree 8.

Mean normalized gap on unseen10000, train size 50. Degree is used here as label-complexity.

Why Compare KEP with Shortest Path?

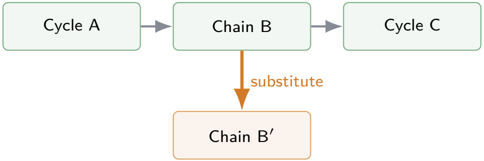
Shortest path

- Serial and coupled decision structure.
- One wrong segment can redirect the whole path.
- Ranking errors can create large regret.
- Standard benchmark where DFL often helps strongly.



Kidney exchange

- Packing of cycles and chains.
- Several exchanges can be substituted.
- Different plans may have similar value.
- DFL gains may be naturally smaller on average.



Puzzle

Why does DFL improve shortest path substantially, but only moderately in KEP?

Key Hypothesis: Close Substitute Solutions

Hypothesis

Kidney exchange often contains many **close substitute exchange plans**.

Why KEP can behave this way

- A feasible plan is a packing of cycles and chains.
- One exchange can often be replaced by another.
- The plan identity changes, but its objective value may stay close.

Oracle plan — Cycle A + Chain B + Cycle C

small replacement

Two-stage plan — Cycle A + Chain B′ + Cycle C

Simple implication

$$\mathrm{gap}(y_{\mathrm{2stage}}) \le \epsilon$$

$$\implies \text{max possible improvement} \le \epsilon.$$

Interpretation

DFL can only remove regret that is actually present.

Look Beyond Rank 1

Diagnostic idea

If KEP has close substitutes, do not only inspect the selected solution.

For each trained model and graph:

1. solve for the best predicted KEP solution;
2. exclude it with a no-good cut;
3. re-solve for rank 2;
4. evaluate both under true reward.

Question

Does solution quality collapse after rank 1, or are alternatives still near-oracle?

Predicted candidate ranking

rank 1 selected by model

rank 2 next distinct alternative → true value

rank 3 another candidate

rank 20 deeper candidate basin

Finding 1: Many Second-Best Solutions Are Still Good

2stage rank 2

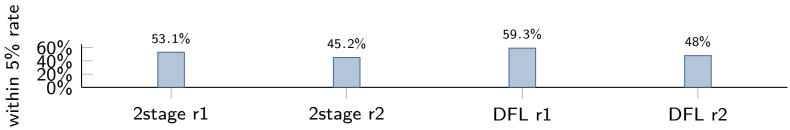
7.04%
mean normalized oracle gap

45.19%
within 5% of oracle

DFL rank 2

6.32%
mean normalized oracle gap

47.97%
within 5% of oracle



Interpretation

Excluding the best predicted solution often still leaves a near-oracle alternative.

The Close-Substitute Pattern Persists

Cycle-length sensitivity

Setup: allow max cycle length $K = 3, 4, 5$; max chain length fixed.

Rank-2 gap	$K = 3$	$K = 4$	$K = 5$
2stage	7.04%	6.93%	6.88%
DFL	6.32%	6.20%	5.61%

Longer cycles do not make rank-2 solutions systematically farther from the oracle.

Density perturbation

- Add/remove arcs changes the feasible landscape.
- Close alternatives often persist.
- Exact promotion/correction mechanisms become graph-dependent.

G-696
close alternatives

G-392
correction

G-1560
promotion

arc perturbations reshape mechanisms

Safe reading

This is not a simple topology law; it supports a graph-instance / feasible-landscape interpretation.

Finding 2: DFL Helps by Reranking Candidate Solutions

Two-stage predicted ranking

rank 1 A: selected by two-stage

rank 2 · · ·

rank 8 B: near-oracle candidate

rank 20 · · ·

DFL selected ranking

rank 1 B: promoted by DFL

candidate promotion

Successful reranking

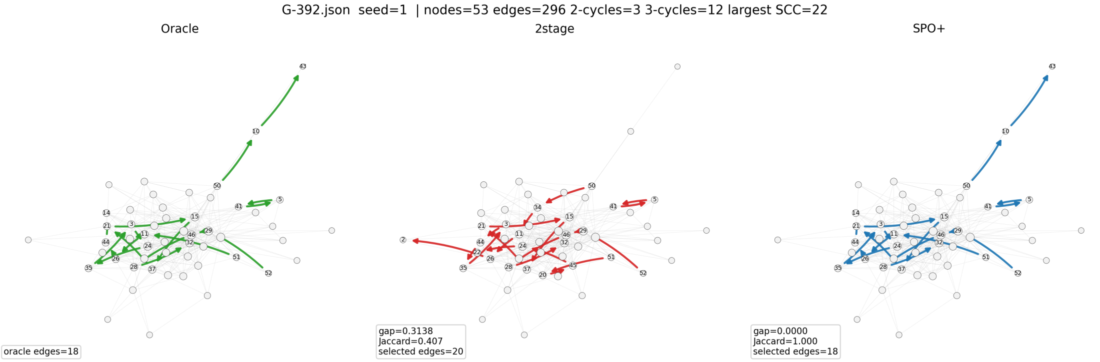
The promoted candidate is near-oracle.

Harmful reranking

The promoted candidate is not near-oracle.

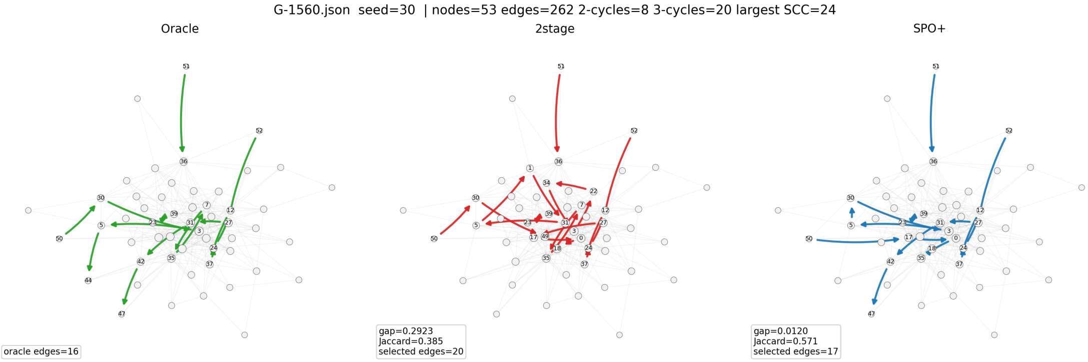
Same mechanism; different promoted-candidate quality.

Case Evidence: G-392 Exact Correction



Two-stage selects a high-gap solution; DFL recovers the oracle solution in this seed.

Case Evidence: G-1560 Near-Oracle Correction



DFL does not exactly match oracle here, but reduces the gap from 0.2923 to 0.0120.

From Case Mechanisms to Graph-Level Diagnostics

Next question

Can graph-level descriptors characterize when DFL is likely to help, hurt, or be unnecessary?

Before prediction
raw topology
feasible-set geometry

prospective

→

After two-stage prediction
ranking boundary
candidate ambiguity

post-training

→

After true-label evaluation
oracle landscape
mechanism audit

retrospective

Evidence boundary

Some diagnostics explain observed mechanisms; not all are deployable pre-match rules.

Topology Alone Does Not Give a Simple Rule

Prospectively available features

- number of vertices and arcs;
- density and degree distribution;
- reciprocity and components;
- 2-cycle / 3-cycle counts;
- feasible exchange and conflict geometry.

Phase 1 readout

- Single-feature Spearman associations are weak.
- No simple rule like "dense graphs are good for DFL."
- Helpful and harmful cases still show graph-instance specificity.

Matched controls

Selected helpful targets remain high relative to topology-matched controls, while harmful controls remain low.

Topology shapes the feasible landscape, but it is not enough by itself.

Graph-Level Signal: Top20 Boundary Marks Opportunity and Risk

Helpful signal

Within-1% top20 count

How many two-stage top20 candidates are within 1% of the predicted rank-1 value?

Helpful AUROC: 0.752

Risk signal

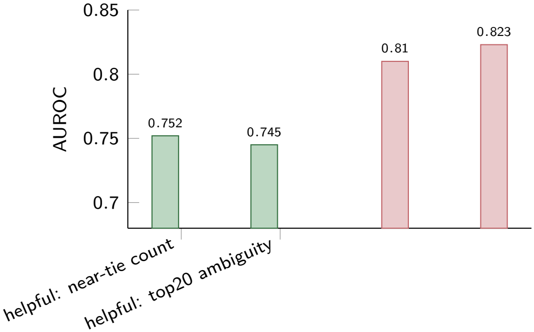
Ambiguity and diversity

The same broad candidate boundary also lights up harmful reranking controls.

Harmful AUROC: 0.810–0.823

Audit scope

160 top20-covered graphs, 50 seeds each, selected targets plus matched controls.



Caution

Top20 ambiguity means DFL reranking can matter; it does not say the reranking will be beneficial.

Takeaways from this Project

1. Decision, not prediction

We are not only fitting edge rewards. We study how prediction errors change downstream exchange decisions.

2. Real KEP observability

Real deployments cannot use retrospective oracle labels. We must separate topology, post-training ambiguity, and after-the-fact mechanism evidence.

3. DFL is conditional

DFL can improve KEP decisions, but its effect is limited when two-stage already selects close substitutes.

Conclusion: DFL for KEP should be mechanism-aware: useful when it promotes near-oracle candidates, limited when close substitutes already exist, and risky when it promotes the wrong candidate.

Next step: tell a good story, turn this into a conference paper.